Uniform quasipolynomial-time trace reconstruction

Abstract

A deletion trace of a binary string x is obtained by keeping each position independently with probability p and deleting it otherwise, then writing the kept bits in order. The kept bits keep no record of where they came from. *Trace reconstruction* asks: given several independent traces of an unknown x, how many traces are needed, and how fast can x be recovered? We give one explicit algorithm that works for every string x, for every known rational retention probability p, with success probability at least $2/3$. If p is held fixed, both the number of traces and the running time are quasipolynomial in the string length n (i.e. of the form $\exp(\mathrm{polylog}(n))$). The bound is also stated uniformly in p: the running time is a fixed polynomial in the number of traces the theorem allows. When the deletion probability is at most $n^{-\varepsilon}$ for a fixed $\varepsilon > 0$, everything becomes polynomial. This document proves that theorem completely.

1 Introduction

The problem in one paragraph

Fix a binary string $x = x_1 \cdots x_n \in \{0,1\}^n$. A *deletion trace* is the random string produced by the following experiment: walk through the positions $1, 2, \ldots, n$; keep position i with probability p and drop it with probability $q = 1 - p$, independently at each i; the kept bits, in their original order, form the trace. For example, if $x = 1011$ and the deletion removes positions 2 and 4, the resulting trace is 11. The trace reports its own length and its bits, but nothing about which original positions survived. An algorithm is handed n, the value p (as a rational number), and a stream of independent traces of x, and must output x. We want an algorithm that succeeds with probability at least $2/3$ *for every* x (a worst-case guarantee), and we want to count the two resources: how many traces it reads, and how many bit operations it performs.

Why the problem is subtle

If the trace were the input up to independent bit flips, we could denoise one coordinate at a time. Here the trouble is different: the same trace rank can come from many different input positions as the random deletions shift everything. So the positions we want are "smeared out", and local statistics do not suffice. What saves us is a family of carefully chosen *nonlocal* statistics (products of bits at ranks that may be far apart) together with a way to test, bit by bit, whether a partial guess is consistent with the data.

Main result

Let $n \geq 2$ and let $p = a/b \in (0,1]$ be a rational in lowest terms. Write L for the total number of bits in a and b when written in binary, so L is the length of the honest input to an algorithm. Write

$q = 1 - p$. All logarithms are natural unless a base is shown. Define the *budget proxy*

$$\mathsf{X}(n,p) = \begin{cases} 1 + \dfrac{\log n}{1 + \log(1/q)}, & q > 0, \\ 1, & q = 0, \end{cases} \tag{1.1}$$

$$B(n,p) = p^{-1} (\log n)\, \mathsf{X}(n,p)^2 \left(1 + \log \mathsf{X}(n,p)\right)^6.$$

Theorem 1.1 (Uniform trace reconstruction)**.** *There are absolute constants $C, c \geq 1$ and one randomized multitape Turing machine $\mathcal{A}$ with the following property. Take any $n \geq 2$, any rational $p \in (0,1]$, and any $x \in \{0,1\}^n$. Given n and p in binary and access to independent deletion traces of x with retention probability p, the machine*

(i) reads at most $M_C(n,p) = \lceil \exp(C\, B(n,p)) \rceil$ traces,

(ii) outputs a string in $\{0,1\}^n$, and

(iii) returns x with probability at least $2/3$.

Moreover, for every *internal sequence of random bits and* every *stream of well-formed traces of strings of length at most n, the machine halts within*

$$\left(n + L + M_C(n,p)\right)^c$$

bit operations. This count already includes reading every trace bit and every delimiter, all preprocessing, all finite-precision arithmetic, and writing the output. The only thing not counted is the external generation of the traces.

Reading the bound

For a fixed p, plugging $q > 0$ constant into (1.1) gives $\mathsf{X}(n,p) = O(\log n)$ and hence $B(n,p) = O_p((\log n)^3(1 + \log\log n)^6)$. Since $M_C = \exp(CB)$, the number of traces is $\exp(O_p((\log n)^3(1 + \log\log n)^6))$, which is quasipolynomial in n. The running time bound $(n + L + M_C)^c$ is then also quasipolynomial. So both resources are quasipolynomial for fixed p. When instead $q \leq n^{-\kappa}$ for a fixed $\kappa > 0$, one has $1 + \log(1/q) \geq \kappa \log n$, so $\mathsf{X}(n,p) \leq 1 + 1/\kappa$ is bounded and p^{-1} is bounded by a constant depending only on κ; then $B(n,p) = O_\kappa(\log n)$ and everything is polynomial in n.

How the proof is organised

The proof is one long chain. We first build the object the decoder tests against: a family of "positive moment functionals", a convex relaxation of the set of genuine probability laws on strings. The heart of the argument (Section 5) is a *separation theorem*: any positive moment functional that insists on a wrong bit at some position must disagree, by an exponentially small but nonzero amount, with one of the observable trace statistics. This is an analytic fact, proved by moving a local signal out to large distance scales (Section 4 and 4.4) and then rotating it into the trace generating function (Section 5). The decoder (Section 6) turns separation into an algorithm: it tests one bit at a time using exact rational feasibility, and we bound the cost. Section 2 records the lineage of these ideas.

2 History and relation to earlier work

Reconstructing a sequence from corrupted copies has a long information-theoretic history. Levenshtein studied reconstruction under several combinatorial and probabilistic channel models [15].

Batu, Kannan, Khanna, and McGregor introduced reconstruction from independent random deletion traces into theoretical computer science, motivated partly by sequence alignment [2]. What makes the deletion channel different from coordinatewise denoising is that a single trace rank can come from many input positions.

For a fixed deletion probability, Holenstein, Mitzenmacher, Panigrahy, and Wieder proved a worst-case sample bound of $\exp(\widetilde{O}(\sqrt{n}))$ [13]. In independent STOC 2017 papers, De, O'Donnell, and Servedio and, separately, Nazarov and Peres proved $\exp(O(n^{1/3}))$ bounds together with matching limitations for methods that use only trace means [11, 17]. Their complex-analytic approach ties deletion statistics to bounded-coefficient polynomials; analytic inequalities of Borwein and Erdélyi are in the background of this method [3]. That mean-based lower bound restricts a method, not arbitrary trace algorithms.

Chase improved the general sample upper bound to $\exp(O(n^{1/5}\log^5 n))$ [6]. On the lower-bound side, Holden and Lyons proved an almost-$n^{5/4}$ bound and Chase strengthened it to $\Omega(n^{3/2}/\log^7 n)$ for fixed deletion probability [12, 5]. Burudgunte, Valiant, and Wang then proved the sample bound $\exp(p^{-7/3}(\log_2 n)^{c_0})$ for an absolute c_0, using nonlocal low-order statistics and a multiscale propagation argument [4, Theorem 43]. Their bound is quasipolynomial when p^{-1} is polylogarithmic in n. Theorem 1.1 builds on this strategy and provides an explicit uniform reconstruction algorithm with the displayed bit-complexity bound.

Sample complexity and reconstruction time need different arguments. At fixed confidence, maximum likelihood reconstructs worst-case using at most $O(n)$ times the optimal sample count, but evaluating it over all candidate strings requires a large search [10, Corollary 1.1 of the full version]. Burudgunte, Valiant, and Wang explicitly separate testing two known strings from reconstructing an unknown string, and leave quasipolynomial-time reconstruction open for their approach [4, Section 1]. The extra separation proved here concerns *every* positive semidefinite moment completion of a wrong prefix. It enables constructive feasibility tests in place of a search over complete strings. Earlier polynomial-time results already cover low-deletion regimes [8].

The use of nonlocal statistics is essential to compare with known barriers. Chen, De, Lee, and Servedio proved strong lower bounds for statistical queries restricted to short *consecutive* trace blocks [7]. Our statistics involve few ranks but may span the whole trace, as in [4]. The generating-function identity used to transfer them from input coordinates has antecedents in higher-order trace reconstruction [9, 6].

The positive moment relaxation is a Boolean instance of the semidefinite moment and sum-of-squares framework [14, 18]. The finite feasibility iteration is related to the classical relaxation methods of Agmon and Motzkin–Schoenberg [1, 16]. We prove the analytic separation and the rational implementation directly; no external trace-reconstruction or optimization theorem is used.

3 Moment relaxations and prefix separation

We now describe exactly what the decoder measures and exactly what convex family it tests against. Everything the algorithm ever sees is a *trace statistic*: an expected product of a few trace ranks. Because we cannot know which input positions produced those ranks, the algorithm treats these statistics as a system of linear equations with a semidefinite (positivity) condition, and solves for a *completion* of a partial guess. This section sets up that relaxation and states the separation theorem that makes the relaxation useful.

Throughout, $[n] = \{1, \dots, n\}$ and $X_1, \dots, X_n$ are formal Boolean variables standing for the unknown bits. Products are always reduced by $X_i^2 = X_i$ (a bit times itself is itself), so every polynomial has degree at most the number of distinct variables it mentions.

3.1 Observable trace moments

Fix a nonempty set of ranks $I = \{\ell_1 < \cdots < \ell_k\} \subseteq [n]$. The corresponding *trace observable* is the product of the bits that sit at those ranks in a single trace; if the trace is shorter than ℓ_k, the observable is 0. Let $T_I(X)$ be its expectation over the deletion channel, viewed as a polynomial in the input bits:

$$T_I(X) = \sum_{i_1<\cdots<i_k} c_{I;\,i_1,\ldots,i_k}\, X_{i_1} \cdots X_{i_k}. \tag{3.1}$$

The coefficient $c_{I;\,i_1,\ldots,i_k}$ is the probability that the input positions $i_1, \ldots, i_k$ land at trace ranks $\ell_1, \ldots, \ell_k$. As the input tuple varies these events are disjoint, so

$$c_{I;\,i_1,\ldots,i_k} \geq 0, \qquad \sum_{i_1<\cdots<i_k} c_{I;\,i_1,\ldots,i_k} \leq 1. \tag{3.2}$$

Thus T_I has degree at most k, nonnegative coefficients, and coefficient sum at most one; its coefficients are rational when p is rational. The algorithm estimates $T_I(x)$ by averaging the observable over independent traces. It never learns the coefficients' input positions from the data, only the numeric value.

3.2 Positive moment arrays

The following Boolean moment relaxation is a special case of the semidefinite moment and sum-of-squares frameworks of Lasserre [14, Section 2] and Parrilo [18, Sections 4.2–4.3]. We use only the defining positivity condition and its elementary consequences, all proved below.

Fix a proposed prefix $u \in \{0,1\}^d$ and a degree bound $D \geq 1$. On the remaining positions $U = \{d+1, \ldots, n\}$ introduce sign variables $Y_i = 2X_i - 1$, so that $Y_i^2 = 1$ and $X_i = (1+Y_i)/2$. A *moment array* assigns real numbers m_J to subsets $J \subseteq U$ with $|J| \leq 2D$, with $m_\varnothing = 1$, subject to the single positive-semidefiniteness condition

$$\mathcal{M}_D(m) = (m_{I \triangle J})_{\substack{I,J \subseteq U \\ |I|,|J| \leq D}} \succeq 0, \tag{3.3}$$

where $\triangle$ is symmetric difference. Only subsets that actually exist are indexed, so the definition still makes sense when $D > |U|$ or U is empty.

The array defines a functional $\mathcal{E}$ on polynomials of degree at most $2D$ as follows: substitute $X_i = u_i$ for $i \leq d$; expand each remaining bit as $(1+Y_i)/2$; reduce $Y_i^2 = 1$; then set $\mathcal{E}[\prod_{i \in J} Y_i] = m_J$ and extend complex-linearly. Call $u \mapsto \mathcal{E}$ a *positive moment functional of order* $2D$ *fixing* u. The word "positive" asserts (3.3), not the existence of an actual probability law realising all these moments; any feasible array, even a fictitious one, is allowed in the test.

Lemma 3.1 (Elementary moment bounds)**.** *For every array obeying* (3.3)*:*

(a) $|m_J| \leq 1$ *for every* $|J| \leq 2D$*;*

(b) under $\mathcal{E}$*, the value of any product of at most* $2D$ *distinct input bits has magnitude at most one;*

(c) on polynomials of degree at most D*, the form* $\langle P, R \rangle_{\mathcal{E}} = \mathcal{E}[P\,\overline{R}]$ *is a positive semidefinite Hermitian form, linear in its first argument; it satisfies Cauchy–Schwarz and becomes an inner product after quotienting by its zero-norm subspace.*

Proof. The empty moment equals one. Any nonempty J with $|J| \leq 2D$ splits as $J = I \sqcup K$ with $|I|, |K| \leq D$. In (3.3) the I, K principal submatrix has diagonal entries one and off-diagonal entry

m_J; positive semidefiniteness of a real symmetric matrix with unit diagonal forces $|m_J| \le 1$. This is (a). For (b), write each unfixed bit as $(1+Y_i)/2$; a product of t unfixed bits becomes $2^{-t}\prod(1+Y_i)$, a convex combination of sign monomials of the Y_i, each of modulus at most one by (a); fixed bits simply remove factors or send the product to zero. For (c), the Gram-form reading of the real symmetric positive semidefinite matrix (3.3), extended to complex coefficient vectors, gives the stated form; Cauchy–Schwarz follows by applying positivity to $P + zR$ for $z \in \mathbb{C}$, and it shows every zero-norm vector is orthogonal to every vector, so the quotient is well defined. □

Now fix the true input $x \in \{0,1\}^n$ and a prefix u of length d that is wrong exactly at its last position,

$$u_i = x_i \ \ (i < d), \qquad u_d = 1 - x_d. \tag{3.4}$$

For any positive moment functional fixing u, define the discrepancy

$$\Delta[P] = \mathcal{E}[P] - P(x). \tag{3.5}$$

Because u agrees with x before d, one has $\Delta[X_d] = \pm 1$; because every monomial using only positions strictly before d agrees with x there, $\Delta[P] = 0$ whenever P mentions only positions $< d$. If $P = \sum_A c_A \prod_{i\in A} X_i$ has degree at most $2D$, then point (b) of Lemma 3.1 gives the useful bound

$$|\Delta[P]| \le 2\sum_A |c_A|. \tag{3.6}$$

For $p < 1$ introduce the bookkeeping quantities

$$q_* = \max(q,\, 1/n), \qquad Q = 1 + \log(1/q_*), \qquad Y = 1 + \frac{\log n}{Q}. \tag{3.7}$$

When $q \ge 1/n$ these give $Y = \mathsf{X}(n,p)$ exactly. When $q < 1/n$ they give $1 \le \mathsf{X}(n,p) \le Y < 2$. In all cases

$$\mathsf{X}(n,p) \le Y \le 2\,\mathsf{X}(n,p), \qquad QY = Q + \log n \le C \log n \tag{3.8}$$

for an absolute constant C, since $n \ge 2$.

Theorem 3.2 (Separation from a wrong prefix)**.** *There exist absolute constants $C_{\mathrm{sep}} \ge 1$ and an integer $H \ge 1$ with the following property. For $n \ge 2$ and $0 < p < 1$ define*

$$D_* = 4^{\lceil \log_2 Y \rceil + H} \le 4^{H+1} Y^2$$

using (3.7). Take any $x \in \{0,1\}^n$, any $d \in [n]$, and any positive moment functional of order $2D$ fixing $x_1, \dots, x_{d-1}, 1 - x_d$ with $D \ge D_*$. Then some nonempty $I \subseteq [n]$ with $|I| \le D_*$ satisfies

$$\bigl|\mathcal{E}[T_I] - T_I(x)\bigr| > \exp\bigl(-C_{\mathrm{sep}}\, B(n,p)\bigr).$$

The theorem holds for *every* feasible array, including arrays that no distribution realises. Its proof occupies Sections 4–5. Once it is available, accurate empirical trace moments rule out every wrong extension of any prefix, because a wrong prefix would force a feasible array that violates such a statistic. Section 6 shows how to actually find the feasible extension using a bounded rational computation.

3.3 Exponentially weighted tuple statistics

The separation proof is analytic, so we set up the objects it uses. Fix $x, d, \mathcal{E}$ as above. For $1 \le k \le 2D$ an *order-k feature* is a family of polynomials indexed by the last position j,

$$f_j = \sum_{i_1<\cdots<i_k=j} X_{i_1}\cdots X_{i_k} \prod_{h=1}^{k-1} z_h^{\,i_{h+1}-i_h}, \qquad j \in [n], \tag{3.9}$$

where $z_h = \exp(-a_h + \mathrm{i}\theta_h)$ with $a_h > 0$ and $\theta_h \in \mathbb{R}$. For $k = 1$ this is just $f_j = X_j$. Each f_j sums over increasing tuples of input positions with exponential weights on the gaps. Put

$$H_f = \prod_{h=1}^{k-1} \Big(1 + \frac{1}{a_h}\Big). \tag{3.10}$$

Summing geometric series over the positive gaps and using $(1-e^{-a})^{-1} \le 1 + 1/a$ shows that $H_f \ge 1$ bounds the sum of the absolute coefficients of every f_j.

For $s > 0$ and $\omega \in \mathbb{R}$, the *signal* of the feature is the exponentially damped Fourier sum

$$\mathcal{S}_f(s,\omega) = \sum_{j=1}^{n} \Delta[f_j]\, e^{(-s+\mathrm{i}\omega)(j-d)}. \tag{3.11}$$

Every monomial in f_j uses positions at most j, so $\Delta[f_j] = 0$ for $j < d$; hence, despite the exponent being centred at d, only $j \ge d$ contribute. For the base case $k = 1$, where $f_j = X_j$, this reads $\mathcal{S}_f(s,0) = \sum_{j\ge d} \Delta[X_j]\, e^{-s(j-d)}$, whose first term is $\Delta[X_d] = \pm 1$; that single term is what makes the starting signal about $1/2$ at decay $s = 4$, as made precise in (5.6). The whole multiscale argument (the next two sections) has one goal: to preserve a detectable signal $\mathcal{S}_f(s,\omega)$ while shrinking both s and $|\omega|$ to $O(1/n)$, at the cost of raising the order k by at most a factor of four per step.

4 Moving a signal to a larger scale

The signal (3.11) is a damped sum: it heavily discounts positions more than about $1/s$ beyond the first disagreement. The next proposition moves a detectable signal to a much smaller decay parameter S', at the price of letting the order grow by at most a factor of four. The overall loss in the logarithm of the signal is kept below a factor of eight. Controlling the *angles* of the new gap parameters is equally important: those angle bounds later let us pass from input coordinates to trace coordinates. Enlarging distance scales while raising the order follows the strategy of Burudgunte, Valiant, and Wang [4], but here the comparison functional may be a truncated positive moment functional, and the scale step uses Gram matrices of polynomial vectors.

Proposition 4.1 (One multiscale step)**.** *There are absolute constants $0 < \beta, \eta_0 < 1$ and $C_0, C_1 \ge 1$ with this property. Let f be an order-k feature from (3.9), and let Δ come from a moment functional of order $2D$ satisfying the prefix conditions of Section 3. Suppose*

$$0 < S \le 4, \qquad 0 < S' \le \min\{1, \eta_0 S\}, \qquad 4k \le D,$$

and, for some $\beta S \le s \le S$ and $|\omega| \le 2.002s$,

$$|\mathcal{S}_f(s,\omega)| \ge e^{-E}, \qquad E \ge C_0\Big(1 + k\log(2k) + \log H_f + \log\frac{1}{S'}\Big). \tag{4.1}$$

Then there is an order-k' feature f', with $1 \le k' \le 4k$, satisfying

$$|\mathcal{S}_{f'}(s', \omega')| \ge e^{-7.98E}, \qquad \beta S' \le s' \le S', \qquad |\omega'| \le 2.002 s', \tag{4.2}$$

and

$$H_{f'} \le H_f^4 \Big(1 + \frac{1}{\beta S'}\Big)^3. \tag{4.3}$$

Moreover there are $1 \le m \le 4$ lists, each equal to the old parameter list $(z_1, \dots, z_{k-1})$ or its complex conjugate, and at most $m-1$ added parameters, so that: each new inner-gap parameter is a nonempty product using at most one entry from each list and at most one added parameter; and every added parameter $e^{-a+\mathrm{i}\theta}$ obeys

$$a \ge \beta S', \qquad |\theta| \le C_1 S. \tag{4.4}$$

The proof splits in two. First, a wide Gaussian window turns the signal into Fourier sums, and positivity of the moment functional forces a product of at most four such sums to keep a signal while its total frequency is small (this section). Second, expanding that product and deforming Gaussian contours produces one feature with the claimed parameters (Section 4.4).

4.1 A Gaussian window and its logarithmic bound

Throughout the proof of Proposition 4.1 set

$$\sigma = \frac{K\sqrt{E}}{S'}, \qquad W(l) = \exp\Big(-\frac{l^2}{2\sigma^2}\Big), \qquad b = \beta S', \tag{4.5}$$

with K a sufficiently large absolute constant. Constants are chosen in the order: K first, then β small enough, then η_0 small enough, and finally C_0 large enough. The choices of K and β also force the contour estimates later.

We will repeatedly use this consequence of (4.1). For any fixed product of powers of $H_f, 1+\sigma, 1/S', 1/b$, together with a factor $\exp(O(k\log(2k)))$, the logarithm is at most

$$C\Big(1 + k\log(2k) + \log H_f + \log\frac{1}{S'}\Big) + C\log E,$$

with C depending only on constants already fixed. Enlarging C_0 makes this at most any chosen positive multiple of E. Only finitely many such estimates occur, so one final choice of C_0 handles them all.

For real t define the polynomial and its value at x,

$$A(t) = \sum_{j=1}^{n} f_j \, W(j-d) \, e^{\mathrm{i}t(j-d)}, \qquad B_x(t) = A(t)(x). \tag{4.6}$$

The coefficient bound on f_j and the Gaussian sum give

$$|B_x(t)|, \quad |\mathcal{E}[A(t)]|, \quad |\Delta[A(t)]| \le U := C H_f (1+\sigma), \tag{4.7}$$

with C absolute and $U \ge 1$. By prefix cancellation,

$$P(z) := \sum_{j=d}^{n} \Delta[f_j] \, W(j-d) \, z^{j-d} \quad \text{satisfies} \quad P(e^{\mathrm{i}t}) = \Delta[A(t)].$$

We first show P keeps a signal at almost the original decay:

$$|P(e^{-s_1+\mathrm{i}\omega})| \geq \tfrac{1}{2}e^{-E} \quad \text{for some} \quad |s_1/s-1| \leq 10^{-4}. \tag{4.8}$$

Let V be a centred real normal variable with variance σ^{-2}. Since $W(l)\,\mathbf{E}\,e^{Vl}=1$,

$$\mathcal{S}_f(s,\omega) = \mathbf{E}\,P(e^{-s+V+\mathrm{i}\omega}).$$

Write $\delta = 10^{-4}$. Exponential tilting gives

$$W(l)\,\mathbf{E}\big[e^{Vl}\mathbf{1}_{\{|V|>\delta s\}}\big] = \Pr\Big(|N(l/\sigma^2,\sigma^{-2})| > \delta s\Big).$$

For $0 \leq l \leq \delta s\sigma^2/2$ this probability is at most $2e^{-c\delta^2 s^2\sigma^2}$; for larger l use the bound one and sum e^{-sl}. Because $|\Delta[f_j]| \leq 2H_f$, the contribution of $|V| > \delta s$ is at most

$$2H_f(1+1/s)\Big(2e^{-c\delta^2 s^2\sigma^2} + e^{-\delta s^2\sigma^2/2}\Big). \tag{4.9}$$

Here $1/s \leq 1/(\beta S')$ and $s^2\sigma^2 \geq \beta^2 K^2 E/\eta_0^2$. Choose η_0 small and then C_0 large to make (4.9) less than e^{-2E}. Then the integral over $|V| \leq \delta s$ has magnitude at least $e^{-E}-e^{-2E} \geq e^{-E}/2$, so its integrand attains magnitude at least that large. This proves (4.8).

Set

$$\Lambda(t) = \log\frac{U}{|P(e^{\mathrm{i}t})|} \geq 0,$$

with value $+\infty$ at a boundary zero. We claim

$$\int_{\mathbb{R}} \Lambda(t)\,\frac{s_1\,dt}{\pi(s_1^2+(t-\omega)^2)} \leq 1.001E. \tag{4.10}$$

For completeness, the Cauchy Fourier transform is

$$\int_{\mathbb{R}} \frac{e^{\mathrm{i}\xi v}}{\pi(1+v^2)}\,dv = e^{-|\xi|}. \tag{4.11}$$

For $\xi > 0$ close the contour in the upper half-plane and take the residue at i; for $\xi < 0$ use the lower half-plane; $\xi = 0$ is direct. Thus periodizing the density in (4.10) gives the circle Poisson measure at $e^{-s_1+\mathrm{i}\omega}$.

To apply it, factor the nonzero polynomial P into linear factors. A factor with a root outside the unit disk has harmonic logarithmic magnitude inside. For a root a inside the disk, the identity $|e^{\mathrm{i}t}-a| = |1-\overline{a}e^{\mathrm{i}t}|$ and the inequality

$$|1-\overline{a}z|^2 - |z-a|^2 = (1-|a|^2)(1-|z|^2) \geq 0$$

bound its contribution by its Poisson integral. Roots on the circle follow by a radial limit; their logarithmic singularities are integrable. Summing over factors gives

$$\int_{\mathbb{R}} \Lambda(t)\frac{s_1\,dt}{\pi(s_1^2+(t-\omega)^2)} \leq \log U - \log|P(e^{-s_1+\mathrm{i}\omega})| \leq E + \log(2U).$$

The last term is at most $1.001E$ by the choice of C_0, proving the claim.

4.2 Frequencies at prescribed differences

We want a frequency $\lambda \in [-2s_1, 2s_1]$ with a large signal, and, for every prescribed difference in that interval, pairs of frequencies with that difference. The product $A(t)\overline{A(t')}$ carries signed frequency $t - t'$. After normalizing by the evaluations at x, these products will be the interior vectors of a chain from 1 to a normalized copy of $A(\lambda)$, with consecutive frequency labels differing by at most b. Each neighbouring inner product then has signed frequency of magnitude at most b, even though $|\lambda|$ may be much larger. The logarithmic estimate turns the existence of enough pairs into a question about pairing Cauchy mass.

Lemma 4.2 (Cauchy matching)**.** *Let $w(v) = 1/(\pi(1+v^2))$. For every $r \geq 0$ there is a finite positive measure μ_r on $\mathbb{R}^2$, supported on $v - v' = r$, whose two marginals sum to at most $w(v)\,dv$. For $r > 0$ its total mass is*

$$2\mu_r(\mathbb{R}^2) = 1 - \frac{4}{\pi}\arctan(e^{-\pi/r}); \tag{4.12}$$

for $r = 0$ its total mass is $1/2$. In particular (4.12) exceeds 0.737 for $0 < r \leq 2$.

Proof. For $r = 0$ put half of $w(v)\,dv$ on the diagonal. Suppose $r > 0$ and partition the line into chains $v + jr$, $j \in \mathbb{Z}$, with $-r/2 \leq v < r/2$. Write $w_j = w(v + jr)$. On the edge from j to $j - 1$, for $j \geq 1$, put weight

$$e_j = \sum_{m \geq 0} (-1)^m w_{j+m}.$$

These weights are nonnegative because the summands decrease along the tail, and $e_j + e_{j+1} = w_j$, so the edges exhaust the weight at every positive vertex. Build the negative-side edges the same way, from the negative tail toward zero. The only possibly unused weight sits at the central vertex, with value

$$R(v) = \sum_{j \in \mathbb{Z}} (-1)^j w(v + jr).$$

To check no weight is overused, periodize w at period $2r$ and subtract its translate by r. By (4.11), with $\tau = e^{-\pi/r}$, this gives the absolutely convergent Fourier series

$$R(v) = \frac{2}{r} \sum_{m \geq 0} \tau^{2m+1} \cos \frac{(2m+1)\pi v}{r}.$$

Writing $a = \pi v/r$, the sum equals

$$\operatorname{Re} \frac{\tau e^{ia}}{1 - \tau^2 e^{2ia}} = \frac{\tau(1 - \tau^2)\cos a}{1 - 2\tau^2 \cos(2a) + \tau^4} \geq 0,$$

since $|a| \leq \pi/2$. Orient each edge with its larger endpoint first and integrate the edge weights against dv over $[-r/2, r/2)$. The resulting measure has the claimed marginal domination. Its uncovered mass is

$$\int_{-r/2}^{r/2} R(v)\,dv = \frac{4}{\pi} \sum_{m \geq 0} \frac{(-1)^m \tau^{2m+1}}{2m+1} = \frac{4}{\pi} \arctan \tau,$$

proving (4.12). The expression decreases with r and at $r = 2$ exceeds 0.737; for instance $e^{-\pi/2} < 0.209$ together with $\arctan u \leq u - u^3/3 + u^5/5$ for $0 \leq u \leq 1$ gives this numeric bound. □

Lemma 4.3 (Frequencies from the logarithmic bound)**.** *Let $s_1, E > 0$, let $|\omega| \leq 2.003 s_1$, and let $\Lambda : \mathbb{R} \to [0, \infty]$ be measurable satisfying (4.10). There is a real λ with*

$$|\lambda| \leq 2s_1, \qquad \Lambda(\lambda) \leq 2.40E. \tag{4.13}$$

There is also an absolute C_2 such that, for every $|y| \le 2s_1$, some real t, t' satisfy

$$t - t' = y, \qquad |t|, |t'| \le C_2 s_1, \qquad \Lambda(t) + \Lambda(t') \le 2.74E. \tag{4.14}$$

Proof. The Cauchy mass of $[-2s_1, 2s_1]$ is at least

$$\frac{\arctan(2 - 2.003) + \arctan(2 + 2.003)}{\pi} > 0.418.$$

Because $1.001/0.418 < 2.40$, averaging proves (4.13).

For (4.14), apply Lemma 4.2 with $r = |y|/s_1 \le 2$ and substitute $t = \omega + s_1 v$, reversing the orientation if needed. The two marginals sum to at most the Cauchy measure in (4.10), while twice the pair mass exceeds 0.737. Discard pairs with an endpoint outside $[-C_2 s_1, C_2 s_1]$; twice the discarded mass is at most twice the Cauchy tail outside that interval, which tends uniformly to zero as $C_2 \to \infty$ since $|\omega|/s_1 \le 2.003$. Choose absolute C_2 so the retained mass exceeds $0.733/2$. Its integral of $\Lambda(t) + \Lambda(t')$ is at most $1.001E$, so some retained pair has sum at most

$$\frac{2(1.001)}{0.733} E < 2.74E,$$

as required. The diagonal construction gives the same when $y = 0$. □

4.3 A product with small total frequency

The window parameters satisfy $|\omega|/s_1 \le 2.002/0.9999 < 2.003$, so Lemma 4.3 applies to our Λ. Individual frequencies in (4.14) need not be small, but their differences can be prescribed. We use these pairs to build a chain of polynomial vectors whose neighbouring inner products have small total frequency. The following Hilbert-space fact explains why such a chain keeps the needed exponential accuracy.

Lemma 4.4 (Gram chaining)**.** *Let $v_0, \dots, v_J$, $J \ge 1$, lie in a complex Hilbert space whose inner product is linear in its first argument. Suppose $\|v_0\| = 1$ and, for some $\xi \ge 0$,*

$$|\|v_i\|^2 - 1| \le \xi \ (0 \le i \le J), \qquad |\langle v_i, v_{i-1}\rangle - 1| \le \xi \ (1 \le i \le J).$$

Then $|\langle v_J, v_0\rangle - 1| \le 2J^2\xi$.

Proof. Put $h_i = v_i - v_{i-1}$. The hypotheses give

$$\|h_i\| \le 2\sqrt{\xi}, \qquad |\langle h_i, v_{i-1}\rangle| \le 2\xi, \qquad \|v_{i-1} - v_0\| \le 2(i-1)\sqrt{\xi}.$$

So Cauchy–Schwarz gives $|\langle h_i, v_0\rangle| \le 2\xi + 4(i-1)\xi$. Sum over i and use $\langle v_0, v_0\rangle = 1$. □

Choose an absolute C_3 large enough that all frequencies supplied by Lemma 4.3 have magnitude at most $C_3 S$. A product of one to four factors, each of the form $A(t)$ or $\overline{A(t)}$ with $|t| \le C_3 S$, is called *balanced* if the sum of its signed frequencies has magnitude at most b; conjugation flips the sign of a frequency.

Lemma 4.5 (A balanced product keeps a signal)**.** *Some balanced product G satisfies*

$$|\Delta[G]| \ge e^{-7.94E}. \tag{4.15}$$

Proof. Suppose not: every balanced product has signal below $\epsilon = e^{-7.94E}$. The squared modulus $|A(t)|^2$ is one such product. Positivity of $\mathcal{E}$ on $A(t)$ and 1 gives

$$|\mathcal{E}[A(t)]|^2 \le \mathcal{E}[|A(t)|^2] \le |B_x(t)|^2 + \epsilon,$$

hence

$$|\Delta[A(t)]| \le 2|B_x(t)| + \sqrt{\epsilon}. \tag{4.16}$$

At λ from (4.13) this implies

$$|B_x(\lambda)| \ge e^{-2.41E}. \tag{4.17}$$

Indeed $|\Delta[A(\lambda)]| \ge e^{-2.40E}$ whereas $\sqrt{\epsilon} = e^{-3.97E}$; fixed factors are absorbed by the lower bound on E in (4.1). For a pair in (4.14), nonnegativity of Λ gives both signals at least $e^{-2.74E}$, and

$$|\Delta[A(t)]\Delta[A(t')]| = U^2 e^{-\Lambda(t)-\Lambda(t')} \ge e^{-2.74E}.$$

Applying (4.16) to each gives

$$|B_x(t)\overline{B_x(t')}| \ge e^{-2.75E}. \tag{4.18}$$

Each evaluation is at least a third of its signal once E is large; the factor $1/9$ is absorbed in $e^{-0.01E}$.

Take a monotone chain $0 = y_0, y_1, \dots, y_J = \lambda$ with consecutive distances at most b and

$$1 \le J \le C(1 + S/b).$$

If $\lambda = 0$, use two endpoints and $J = 1$. For each interior y_i choose a pair from (4.14) and set

$$V_0 = 1, \qquad V_i = \frac{A(t)\overline{A(t')}}{B_x(t)\overline{B_x(t')}} \ (0 < i < J), \qquad V_J = \frac{A(\lambda)}{B_x(\lambda)}.$$

The denominators are nonzero by (4.17)–(4.18), and every $V_i(x) = 1$. View these polynomials as vectors v_i in the Hilbert space of $\mathcal{E}$; they have degree at most $2k \le D$.

For $1 \le i \le J$ the norm square of v_i and its inner product with v_{i-1} are expectations of normalized balanced products, whose signed frequencies are respectively zero and $y_i - y_{i-1}$; each product has at most four factors and its normalizing denominator has magnitude at least $e^{-5.50E}$. The contradictory assumption therefore gives

$$|\|v_i\|^2 - 1| \le \xi, \qquad |\langle v_i, v_{i-1}\rangle - 1| \le \xi, \qquad \xi = \epsilon e^{5.50E} = e^{-2.44E}.$$

Since $v_0 = 1$ has norm one, Lemma 4.4 yields

$$|\Delta[A(\lambda)]| = |B_x(\lambda)|\,|\langle v_J, v_0\rangle - 1| \le 2UJ^2 e^{-2.44E} < e^{-2.40E}.$$

The last strict inequality holds after enlarging C_0, because $\log(2UJ^2) < 0.04E$ by (4.1). This contradicts (4.13) and proves the lemma. □

We now have a product of at most four windowed feature sums with signal at least $e^{-7.94E}$ and total frequency at most b. It remains to turn it into one feature while keeping both decay and frequency of order S'; the Gaussian contour construction in the next section does exactly that.

4.4 Converting a balanced product to a feature

Lemma 4.5 gives a product of at most four windowed Fourier sums with a substantial discrepancy and small total frequency. We now write that product as an integral of features. The contour puts their outer parameters in the sector that Proposition 4.1 requires; what remains is to bound the damping on the inner gaps.

Fix a product from Lemma 4.5 and let $1 \le m \le 4$ be its number of factors. Expand each factor into the tuples that define f. An *interleaving* is an ordering, ties included, of the mk labeled positions in these tuples. There are at most $(mk)^{mk} = \exp(O(k\log(2k)))$ interleavings: assign each labeled position its rank among the distinct positions. Keep only interleavings that increase strictly inside each old tuple. Fix one interleaving and relabel its copies so $j_1 \le \cdots \le j_m$. Ties between endpoints may be ordered arbitrarily. Absorbing conjugation into the signs of the frequencies, the window weight is

$$\prod_{r=1}^{m} W(j_r - d)\exp\bigl(\mathrm{i}t_r(j_r - d)\bigr), \qquad |t_r| \le C_3 S, \qquad \Bigl|\sum_{r=1}^{m} t_r\Bigr| \le b. \tag{4.19}$$

Conjugation also conjugates the old inner-gap parameters, leaving their dampings unchanged.

A Gaussian contour with controlled total frequency. Recall $\sigma = K\sqrt{E}/S'$ and $b = \beta S'$. For every real l,

$$W(l) = \frac{\sigma}{\mathrm{i}\sqrt{2\pi}} \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} \exp\Bigl(\frac{\sigma^2 v^2}{2} + vl\Bigr)\, dv. \tag{4.20}$$

Indeed, $v = \mathrm{i}u$ gives the Gaussian Fourier transform; its value at $l = 0$ is one, and integration by parts shows its l-derivative is $-l/\sigma^2$ times itself, fixing the formula.

Set $\rho = 1/2.002$. For $u = (u_1, \dots, u_m) \in \mathbb{R}^m$ define

$$\Omega = \sum_{r=1}^{m}(u_r + t_r), \qquad v_r = b + \mathrm{i}u_r \ (r < m), \qquad v_m = -mb - \rho|\Omega| + \mathrm{i}u_m. \tag{4.21}$$

These real parts have partial sums rb for $r < m$ and total $-b - \rho|\Omega|$. Centred at j_m, the total sum gives the real part of the outer coefficient and the negatives of the partial sums give the real parts on consecutive endpoint gaps. So the contour is built to produce outer decay $b + \rho|\Omega|$ and positive inner dampings rb; the exact rewriting is checked below.

For each fixed tuple of positions, first shift the first $m - 1$ contours of (4.20) to the vertical lines $\operatorname{Re} v_r = b$, leaving the last on the imaginary axis. The linking horizontal integrals vanish by Gaussian decay. Next fix $(u_1, \dots, u_{m-1})$ and deform the last contour to the piecewise-linear graph in (4.21), whose slopes have absolute value $\rho < 1$. Along horizontal segments that join its truncations to the imaginary axis at height $\pm T$, the Gaussian exponent has real part at most $-cT^2 + O(T)$ for some $c > 0$, while $\exp(v_m(j_m - d))$ adds only a linear term in T, so those linking integrals also vanish. The integrands are entire, so no residues occur. This proves the identities as iterated integrals; joint absolute convergence follows from the bound below.

Let ν be the resulting complex measure on $\mathbb{R}^m$ after removing the position-dependent factors. Outside the measure-zero set $\Omega = 0$,

$$d\nu(u) = \Bigl(\frac{\sigma}{\sqrt{2\pi}}\Bigr)^m \exp\Bigl(\frac{\sigma^2}{2}\sum_{r=1}^{m} v_r^2\Bigr)\bigl(1 + \mathrm{i}\rho \operatorname{sgn}\Omega\bigr)\, du. \tag{4.22}$$

The last factor is the differential of the bent contour divided by $\mathrm{i}\, du_m$. We now bound this uniformly over the frequencies in (4.19); the bounds also give absolute convergence.

Put $\delta_0 = 1 - 4\rho^2 > 0$ and $c_1 = \delta_0/2$. Since $m \le 4$ and $|\sum t_r| \le b$, Young's inequality gives an absolute C_4 with

$$\begin{aligned}\sum_{r=1}^{m} (\operatorname{Re} v_r)^2 &= (m-1)b^2 + (mb + \rho|\Omega|)^2 \\ &\le 3b^2 + \Big(2\rho\sqrt{\textstyle\sum_r u_r^2} + (4+\rho)b\Big)^2 \\ &\le (1-c_1)\sum_r u_r^2 + C_4 b^2. \end{aligned} \tag{4.23}$$

The strict inequality $4\rho^2 < 1$ is essential: summing at most four imaginary coordinates can multiply their squared norm by four, and the sector constant 2.002 leaves a positive Gaussian margin. For example $C_4 = 3 + (4+\rho)^2(1 + 8\rho^2/\delta_0)$ works. With (4.22) this gives

$$|d\nu(u)| \le C\sigma^m \exp\Big(\frac{C_4\sigma^2 b^2}{2} - \frac{c_1\sigma^2}{2}\sum_r u_r^2\Big) du.$$

For any fixed tuple the remaining exponential factor has modulus at most $\exp(C_n(1 + \sum_r |u_r|))$ for a finite C_n, which is integrable against the Gaussian bound. So the iterated deformation is a jointly absolutely convergent integral, and finite tuple sums may be exchanged with it.

Write $\mathcal{U} = \{u : \max_r |u_r| \le S'/16\}$. Integrating the last bound and splitting the negative quadratic term into two equal halves on $\mathcal{U}^c$ gives absolute constants C, C' with

$$\begin{aligned} \|\nu\|_{\mathrm{abs}} &\le C\exp(C_4 K^2\beta^2 E/2), \\ \int_{\mathcal{U}^c} |d\nu| &\le C'\exp\Big[\Big(\frac{C_4K^2\beta^2}{2} - \frac{c_1K^2}{1024}\Big)E\Big]. \end{aligned}$$

Here $\|\nu\|_{\mathrm{abs}}$ is total variation; the powers σ^m cancel under Gaussian rescaling. Choose K with $c_1K^2/1024 \ge 30$, then $\beta \le 1/4$ with $C_4K^2\beta^2/2 \le 0.001$. Enlarging C_0 to absorb fixed prefactors gives

$$\|\nu\|_{\mathrm{abs}} \le e^{0.005E}, \qquad \int_{\mathcal{U}^c} |d\nu| \le e^{-20E}. \tag{4.24}$$

This respects the constant order fixed earlier: K first, β second, then η_0 small to meet the window estimate, and only afterward C_0 enlarged.

Outer parameters and endpoint gaps. For $c_r = v_r + \mathrm{i}t_r$, telescoping the endpoint differences gives

$$\sum_{r=1}^{m} c_r(j_r - d) = \Big(\sum_{r=1}^{m} c_r\Big)(j_m - d) - \sum_{r=1}^{m-1}\Big(\sum_{l=1}^{r} c_l\Big)(j_{r+1} - j_r). \tag{4.25}$$

So the largest endpoint has coefficient $-s' + \mathrm{i}\Omega$ with

$$s' = b + \rho|\Omega|. \tag{4.26}$$

The link from j_r to j_{r+1} receives the parameter

$$\exp\Big(-rb - \mathrm{i}\sum_{l=1}^{r}(u_l + t_l)\Big), \qquad 1 \le r < m. \tag{4.27}$$

Thus every added damping is at least b, uniformly in u. On $\mathcal{U}$,

$$|\Omega| \le (\beta + 1/4)S', \qquad \beta S' \le s' \le [\beta + \rho(\beta + 1/4)]S' \le S', \qquad |\Omega| \le 2.002 s'.$$

The angle in (4.27) has absolute value at most $3(C_3S + S'/16) \le C_1 S$ for a fixed absolute C_1. Hence the outer parameters satisfy (4.2) and the endpoint links satisfy (4.4). The choice of C_1 uses only C_3 and $S' \le S$, imposing no new condition on K or β.

4.5 Interleavings and the cost of the inner gaps

We have outer weights at the right scale. To finish, each interleaving must have exactly the form of a feature with controlled coefficient sum. One old inner edge can cross several gaps of the merged tuple, so charging its damping at every crossing would overcount. A distinct-edge assignment prevents that loss.

For the fixed interleaving write its distinct positions as $i_1 < \cdots < i_v$ with $1 \le v \le mk$. Form a multigraph on these vertices: include the inner edges of each old tuple and the links between consecutive endpoints in (4.27), each edge carrying its exponential parameter. It is connected, since each old tuple is a path to its endpoint and the endpoint links join those paths. Delete loops from equal endpoints; their weights are one.

For each merged gap $i_{a+1} - i_a$, multiply the parameters of all edges crossing it. Each edge length is the sum of the merged gaps it crosses, so the new parameters reproduce the original weight exactly. Every merged gap is crossed by some edge, by connectivity, so its damping is positive. Within one old tuple the inner-edge intervals have disjoint interiors, and the same holds for the consecutive endpoint links. It follows that each new parameter uses at most one old parameter from each copy and at most one added endpoint parameter. This is exactly the factor-count assertion of Proposition 4.1.

Ties merge repeated bit variables by $X_i^r = X_i$ for $r \ge 1$. Once the interleaving is fixed, choosing arbitrary strictly increasing values for its v distinct positions specifies exactly one assignment to its labeled tuples, so no further restriction remains on the new tuple. Its largest position is $i_v = j_m$. Summing all these assignments gives an order-v feature $f^{\mathcal{I},u}$ with outer weight $\exp((-s' + \mathrm{i}\Omega)(i_v - d))$.

The needed combinatorial fact controls its height.

Lemma 4.6 (Assigning distinct edges to consecutive gaps)**.** *Let a connected multigraph have ordered vertices $i_1 < \cdots < i_v$. There is an assignment of a distinct edge to each gap (i_a, i_{a+1}) so that the assigned edge crosses that gap.*

Proof. For $v = 1$ there is nothing to prove. Otherwise keep a spanning tree and orient each edge from its smaller to its larger endpoint. Let M be the $(v-1) \times (v-1)$ matrix whose (e, a) entry is one if tree edge e crosses gap a and zero otherwise. It is invertible: for g in its kernel set $y_1 = 0$ and $y_{a+1} = y_a + g_a$; then $Mg = 0$ says y is equal at the endpoints of every tree edge, and connectivity forces all values zero, so $g = 0$. A nonzero term in the determinant expansion of M pairs its rows and columns through entries equal to one; that pairing is the required assignment. □

To see why distinctness matters, look at Figure 1: one edge may contribute to several new parameters, but it need pay for only one of their upper bounds. Concretely, take two old tuples (i_1, i_3, i_5) and (i_2, i_4, i_6) joined by one dashed endpoint link-edge, with gaps $g_a = (i_a, i_{a+1})$ for $1 \le a \le 5$. The assignment $g_a \mapsto e_a$, $1 \le a \le 5$, gives every gap a distinct crossing edge: e_1 crosses both g_1 and g_2 but is charged only to g_1, and e_2 supplies the bound for g_2.

Give each edge occurrence e its damping $a_e > 0$. A merged gap's damping is the sum of a_e over edges crossing it, hence at least that of its assigned edge. Since $1 + 1/a$ decreases with a, Lemma 4.6

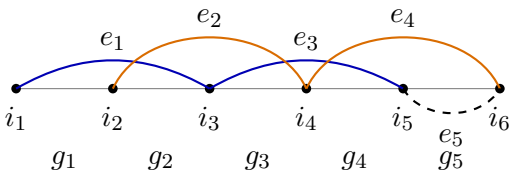


Figure 1: A schematic interleaving of the two old tuples (i_1, i_3, i_5) and (i_2, i_4, i_6), joined by the dashed endpoint link e_5. Here g_a is the gap (i_a, i_{a+1}). The assignment $g_a \mapsto e_a$ for $1 \le a \le 5$ uses a distinct crossing edge for every gap: e_1 crosses both g_1 and g_2 but is charged only to g_1, while e_2 supplies the bound for g_2.

gives

$$\begin{aligned} H_{f^{\mathcal{I},u}} &\le \prod_{e \text{ assigned}} (1 + 1/a_e) \le \prod_{e \text{ present}} (1 + 1/a_e) \\ &\le H_f^m (1 + 1/b)^{m-1} \le H_f^4 (1 + 1/b)^3. \end{aligned} \tag{4.28}$$

The second inequality uses that all factors are at least one; the endpoint-link dampings are $rb \ge b$. This covers all overlaps between old tuples and also the single-distinct-position case, where the height is the empty product one.

Completing the multiscale estimate. Apply Δ to the expanded product and use the integral representation for each interleaving. With $\nu_{\mathcal{I}}$ the corresponding measure,

$$\Delta[\text{product}] = \sum_{\mathcal{I}} \int_{\mathbb{R}^m} \mathcal{S}_{f^{\mathcal{I},u}}(s', \Omega)\, d\nu_{\mathcal{I}}(u). \tag{4.29}$$

All estimates are uniform over interleavings, and the integrands are uniformly bounded even outside $\mathcal{U}$. Indeed, for each fixed endpoint j the coefficients of $f_j^{\mathcal{I},u}$ have absolute sum at most the height in (4.28), terms with $j < d$ vanish under Δ by prefix agreement, and $s' \ge b$ gives

$$|\mathcal{S}_{f^{\mathcal{I},u}}(s', \Omega)| \le 2H_f^4 (1 + 1/b)^3 \sum_{l \ge 0} e^{-bl} \le 2H_f^4 (1 + 1/b)^4. \tag{4.30}$$

No exponential factor from negative distances appears.

Let N_{int} be the number of interleavings and $V = 2H_f^4(1 + 1/b)^4$. Hypothesis (4.1) with large enough absolute C_0 gives

$$E \ge 100, \qquad \log N_{\text{int}} \le 0.01E, \qquad \log V \le 0.01E.$$

Here $\log(1 + 1/b)$ is at most an absolute constant plus $\log(1/S')$, because $b = \beta S'$, $S' \le 1$, and β is already fixed. If every signal in (4.29) with $u \in \mathcal{U}$ had magnitude below $e^{-7.98E}$, then (4.24) and (4.30) would give

$$|\Delta[\text{product}]| \le N_{\text{int}} \big(e^{(0.005 - 7.98)E} + V e^{-20E}\big) \le e^{-7.965E} + e^{-19.98E} < e^{-7.94E}.$$

This contradicts Lemma 4.5. Some interleaving and some $u \in \mathcal{U}$ therefore give a feature with signal at least $e^{-7.98E}$. Its order is at most $mk \le 4k$; (4.26) and the bounds on $\mathcal{U}$ give the outer parameters; (4.28) gives (4.3); and the crossing description gives the parameter factors and (4.4). This completes the proof of Proposition 4.1.

5 From an input signal to separation of trace moments

We now prove Theorem 3.2. Proposition 4.1 keeps a detectable input-coordinate signal while lowering its decay rate. First we choose enough scales to force the rate down to $1/n$. The resulting gap parameters need not lie in the disk where trace moments control the signal, so a single one-variable continuation bridges the gap. Throughout this section $0 < p < 1$ and q_*, Q, Y are as in (3.7). Recall from (3.8) that

$$1 \le \mathsf{X}(n,p) \le Y \le 2\mathsf{X}(n,p), \qquad QY = Q + \log n \le C \log n. \tag{5.1}$$

5.1 A scale schedule and the surviving signal

Set

$$\mu = 7.98, \qquad \alpha = 0.001, \qquad h = \lceil \log_2 Y \rceil + H, \qquad D_* = 4^h, \tag{5.2}$$

where the absolute integer H is fixed below; in particular $D_* \le 4^{H+1}Y^2$. Fix any positive moment functional of order $2D$ with $D \ge D_*$ and first wrong position d, and let Δ be its difference from the true word, as in Section 3. Every constant chosen below is independent of this system, of n, and of p.

For an absolute constant $\eta > 0$ define

$$S_0 = 4, \qquad S_j = \eta\, q_*\, Y^{-\alpha}\, S_{j-1}\, \min\{1, D_* S_{j-1}\} \quad (1 \le j \le h). \tag{5.3}$$

The linear regime of this recurrence pushes D_*S_j below one; its quadratic regime then makes S_j shrink fast. Both regimes also control the arguments of the final gap parameters.

Choose an integer J_* with $\alpha J_* \ge 3$ and set $H = J_* + 3$. To check the decrease, write $\vartheta = \eta q_* Y^{-\alpha}$ and $t_j = D_* S_j$, so $t_j = \vartheta\, t_{j-1} \min\{1, t_{j-1}\}$. Then, whenever the quadratic regime begins,

$$t_{J_*} \le \vartheta^{J_*} t_0 \le 4^{H+2} \eta^{J_*} Y^{2-\alpha J_*} \le 1$$

once η is small enough. Require also $\eta \le \min\{\eta_0, 1/4, e^{-1}\}$, with η_0 from Proposition 4.1. Now $\log(1/\vartheta) \ge Q$ and $t_j \le 1$ for $j \ge J_*$. Iterating $t_j = \vartheta t_{j-1}^2$ in that range gives

$$t_h \le \vartheta^{\,2^{h-J_*}-1} \le \exp\{-Q(2^{h-J_*}-1)\} \le n^{-1}, \qquad S_h \le n^{-1}. \tag{5.4}$$

The last inequality uses $2^{h-J_*} \ge 8Y$ and $Q(Y-1) = \log n$. At each transition $0 < S_j \le \min\{1, \eta_0 S_{j-1}\}$, as the proposition requires.

We also need the cost of these small scales. Put $A_0 = Q + \log Y$. With $\ell_j = \max\{0, \log(1/S_j)\}$ the recurrence gives

$$\ell_j \le 2\ell_{j-1} + \log(1/\vartheta), \quad \ell_0 = 0, \quad \log(1/\vartheta) \le C A_0,$$

hence

$$\log(1/S_j) \le C 2^j A_0 \qquad (0 \le j \le h). \tag{5.5}$$

Now iterate Proposition 4.1. Start with the order-one feature $f_i^{(0)} = X_i$ at decay $s_0 = 4$ and frequency zero. Prefix agreement gives $\Delta[X_i] = 0$ for $i < d$ while $|\Delta[X_d]| = 1$; since $|\Delta[X_i]| \le 2$ for all i,

$$|\mathcal{S}_{f^{(0)}}(4,0)| \ge 1 - 2\sum_{r \ge 1} e^{-4r} = 1 - \frac{2}{e^4 - 1} > \frac{1}{2}. \tag{5.6}$$

Use error exponents

$$E_j = C_5 A_0 \mu^j \qquad (0 \le j \le h), \tag{5.7}$$

with absolute C_5 chosen after H and η.

The induction maintains this statement. At stage j there is a feature $f^{(j)}$ of order $k_j \le 4^j$ with

$$\begin{gathered} \beta S_j \le s_j \le S_j, \quad |\omega_j| \le 2.002 s_j, \quad |\mathcal{S}_{f^{(j)}}(s_j, \omega_j)| \ge e^{-E_j}, \\ \log H_{f^{(j)}} \le C 4^j A_0. \end{gathered} \tag{5.8}$$

The base case follows from (5.6) and $H_{f^{(0)}} = 1$ on increasing C_5. At the transition $j-1 \to j$ the degree requirement is $4k_{j-1} \le 4^j \le D_* \le D$. The proposition's height conclusion and (5.5) give

$$\log H_{f^{(j)}} \le 4 \log H_{f^{(j-1)}} + 3 \log(1 + 1/(\beta S_j)) \le 4 \log H_{f^{(j-1)}} + C 2^j A_0,$$

and summing this proves the height bound in (5.8). For the exponent hypothesis, the proposition's bracket is at most

$$C(1 + j 4^{j-1} + 4^{j-1} A_0 + 2^j A_0).$$

Because $\mu > 4$, the sequences $j(4/\mu)^{j-1}$ and $2^j/\mu^{j-1}$ are bounded. Thus one choice of C_5 makes this bracket times C_0 at most E_{j-1} for every j. All hypotheses of Proposition 4.1 hold, and its signal bound is $e^{-\mu E_{j-1}} = e^{-E_j}$.

Write $f = f^{(h)}$, $k = k_h$, $s = s_h$, $\omega = \omega_h$. We have a signal of magnitude at least e^{-E_h} with $k \le D_*$ and $s \le 1/n$. The next step records where its parameters sit inside the unit disk; this controls the cost of continuing the signal to the trace domain.

5.2 Damping under multiplication

Call the parameters introduced between endpoints in any application of Proposition 4.1 *elementary parameters*, and include the final outer parameter $z_0 = e^{-s+\mathrm{i}\omega}$. An elementary parameter $e^{-a+\mathrm{i}\theta}$ introduced at stage j has $a \ge \beta S_j$ and $|\theta| \le C_1 S_{j-1}$. Using (5.3), pick a small absolute $c_2 > 0$ so that every elementary parameter obeys

$$a \ge \gamma \min\{|\theta|, D_* \theta^2\}, \qquad \gamma = c_2 q_* Y^{-\alpha} \le 1. \tag{5.9}$$

Indeed, the increasing map $u \mapsto u \min\{1, D_* u\}$ changes by at most a factor $\max\{C_1, C_1^2\}$ when u becomes $C_1 u$, so $c_2 \le \beta\eta/\max\{C_1, C_1^2\}$ suffices; shrinking c_2 further covers z_0 since $|\omega| \le 2.002 s$. Conjugating an elementary parameter preserves (5.9).

Each new inner gap holds at most one parameter from each of at most four copies of the preceding feature, and at most one new elementary parameter; count repeated factors with multiplicity, and conjugation does not change the count. Starting with no inner parameters, the maximum number N_j of elementary factors in a stage-j inner gap satisfies

$$N_0 = 0, \qquad N_j \le 4 N_{j-1} + 1, \qquad N_j \le \frac{4^j - 1}{3}.$$

In particular an inner gap of f, even after multiplying by z_0, contains at most $(4^h + 2)/3 \le D_*$ elementary factors.

The next lemma explains why this count matters. The Cayley map $z \mapsto (1+z)/(1-z)$ sends the unit disk to the right half-plane; we need a quantitative distance from its boundary.

Lemma 5.1 (Products with controlled damping)**.** *Let $N \geq 1$ be an integer and $0 < \gamma \leq 1$. Suppose z is a nonempty product of at most N numbers $e^{-a_j+\mathrm{i}\theta_j}$ with $a_j > 0$, $\theta_j \in \mathbb{R}$, and $a_j \geq \gamma \min\{|\theta_j|, N\theta_j^2\}$. Then for some absolute $c_3 > 0$,*

$$\operatorname{Re} \frac{1+z}{1-z} \geq c_3\gamma.$$

Proof. Write $z = e^{-T+\mathrm{i}\psi}$ with $T = \sum_j a_j > 0$ and $\psi = \sum_j \theta_j$. Let

$$U = \sum_{|\theta_j| \geq 1/N} |\theta_j|, \qquad V = \sum_{|\theta_j| < 1/N} |\theta_j|.$$

Summing the hypotheses and applying Cauchy–Schwarz to at most N small angles gives

$$T \geq \gamma\Big(U + N \sum_{|\theta_j|<1/N} \theta_j^2\Big) \geq \gamma(U + V^2).$$

If $U \geq 1$ or $V \geq 1$ the bracket is at least one; otherwise $U + V^2 \geq U^2 + V^2 \geq (U+V)^2/2$. Since $|\psi| \leq U + V$, in all cases

$$T \geq \tfrac{1}{2}\gamma \min\{1, \psi^2\}.$$

For $0 < T \leq 1$ this gives

$$1 - |z|^2 \geq cT, \qquad |1-z|^2 \leq C(T^2 + \min\{1,\psi^2\}) \leq C'T/\gamma.$$

Their ratio is $\operatorname{Re}((1+z)/(1-z))$. If $T > 1$, that ratio is at least $(1-e^{-2})/4$, which also suffices because $\gamma \leq 1$. □

5.3 The exact trace generating identity

Let $z_1, \ldots, z_{k-1}$ be the inner gap parameters of the final feature f. Define the polynomial in k complex variables

$$F(Z_0, \ldots, Z_{k-1}) = \sum_{1 \leq i_1 < \cdots < i_k \leq n} \Delta\Big[\prod_{r=1}^{k} X_{i_r}\Big] Z_0^{i_1} \prod_{r=1}^{k-1} Z_r^{i_{r+1}-i_r}. \tag{5.10}$$

At the point $Z_0^* = z_0$, $Z_r^* = z_r z_0$ $(1 \leq r < k)$, the powers of z_0 telescope to i_k, so

$$F(Z^*) = z_0^d \, \mathcal{S}_f(s,\omega), \qquad |F(Z^*)| \geq e^{-E_h - 1}, \tag{5.11}$$

where $sd \leq nS_h \leq 1$. Each coordinate of Z^* is a nonempty product of at most D_* elementary parameters, so Lemma 5.1 gives

$$\operatorname{Re} \frac{1+Z_r^*}{1-Z_r^*} \geq c_3\gamma \qquad (0 \leq r < k). \tag{5.12}$$

We also need the coordinates' collective closeness to the unit circle. If $z_r = e^{-a_r+\mathrm{i}\theta_r}$, then

$$\frac{1}{1-|Z_r^*|} \leq 1 + \frac{1}{a_r + s} \leq 1 + \frac{1}{a_r} \ (r \geq 1), \qquad \frac{1}{1-|Z_0^*|} \leq 1 + \frac{1}{s},$$

using $e^u \geq 1 + u$. The height bound in (5.8) and (5.5) therefore give

$$W_* := \log \prod_{r=0}^{k-1} \frac{1}{1-|Z_r^*|} \leq \log H_f + \log(1 + 1/(\beta S_h)) \leq C4^h A_0. \tag{5.13}$$

Now express the same polynomial through observable trace moments on a different domain. This is the multivariate trace generating identity of [9, Section 5.2]; see also [7, Fact 20]. We give the counting argument directly. For a rank tuple $\ell = (\ell_1, \dots, \ell_k)$ with $1 \le \ell_1 < \cdots < \ell_k \le n$ write $T_\ell = T_{\{\ell_1,\dots,\ell_k\}}$. For $u = (u_0, \dots, u_{k-1}) \in \mathbb{C}^k$ set

$$G(u) = \sum_{1 \le \ell_1 < \cdots < \ell_k \le n} \Delta[T_\ell]\, u_0^{\ell_1 - 1} \prod_{r=1}^{k-1} u_r^{\ell_{r+1} - \ell_r - 1}.$$

Then, with $Z_r = q + p u_r$, the exact identity is

$$\Big(\prod_{r=0}^{k-1} Z_r\Big) G(u) = p^k F(Z). \tag{5.14}$$

To prove it, fix input positions $i_1 < \cdots < i_k$ that supply the specified retained bits. Their own retention contributes p^k. Every input position before i_1 contributes $q + p u_0$ according to whether it is deleted or retained; there are $i_1 - 1$ such positions. Similarly, the $i_{r+1} - i_r - 1$ positions between i_r and i_{r+1} each contribute $q + p u_r$, and positions after i_k contribute a factor of one. The retained counts in these intervals are exactly $\ell_1 - 1$ and $\ell_{r+1} - \ell_r - 1$. Summing over all input tuples and applying Δ gives $G(u)$ as p^k times the polynomial of (5.10) with every displayed gap exponent reduced by one; multiplying by $\prod_r Z_r$ proves (5.14), including points where some $Z_r = 0$.

Suppose for the moment that

$$|\Delta[T_I]| \le \varepsilon \quad \text{for every } I \subseteq \{1, \dots, n\} \text{ with } 1 \le |I| \le D_*. \tag{5.15}$$

Since $k \le D_*$ the rank sum has at most n^k terms, so $|G(u)| \le n^k \varepsilon$ when all $|u_r| \le 1$; on that polydisk $|Z_r| \le q + p = 1$. Thus (5.14) gives

$$|F(Z)| \le (n/p)^k \varepsilon \qquad \text{whenever } |Z_r - q| \le p \ (0 \le r < k). \tag{5.16}$$

This uses only ordinary trace ranks and the convention that a rank product is zero when the trace is too short.

5.4 Simultaneous continuation and the separation bound

The lower bound (5.11) and the upper bound (5.16) live at different points. In Cayley coordinates the disk $|Z - q| \le p$ becomes a half-plane:

$$|Z - q| \le p \iff \operatorname{Re} \frac{1 + Z}{1 - Z} \ge \frac{q}{p} \qquad (|Z| < 1). \tag{5.17}$$

Indeed, writing $w = (1 + Z)/(1 - Z)$ and expanding $|pw - (1 + q)|^2 \le p^2 |w + 1|^2$ reduces it to $4q \le 4p \operatorname{Re} w$. We move all coordinates into this half-plane with one complex variable. A single strip interpolation avoids paying a separate continuation loss per coordinate.

Write

$$w_r = \frac{1 + Z_r^*}{1 - Z_r^*} = \xi_r + \mathrm{i}\zeta_r, \qquad \xi_r \ge c_3 \gamma,$$

with $\xi_r, \zeta_r \in \mathbb{R}$. For $v \in \mathbb{C}$ set

$$w_r(v) = (1 + v)\xi_r + \mathrm{i}\zeta_r, \quad Z_r(v) = \frac{w_r(v) - 1}{w_r(v) + 1}, \quad R = 1 + \frac{q}{p c_3 \gamma}. \tag{5.18}$$

A common real value of v moves every starting coordinate into the trace half-plane at once. Concretely, in the Z-plane the region $|Z-q| \le p$ is a disk; after the Cayley map $w = (1+Z)/(1-Z)$ it becomes the half-plane $\operatorname{Re} w \ge q/p$. Each path $w_r(v) = (1+v)\operatorname{Re} w_r + \mathrm{i}\operatorname{Im} w_r$ is a horizontal segment starting at w_r and ending at $w_r(R)$ with $\operatorname{Re} w_r(R) = (1+R)\xi_r \ge q/p$. All coordinates use the same parameter $0 \le v \le R$; the proof interpolates $F(Z_0(v), \dots, Z_{k-1}(v))$ on one complex strip, simultaneously for every coordinate (Figure 2).

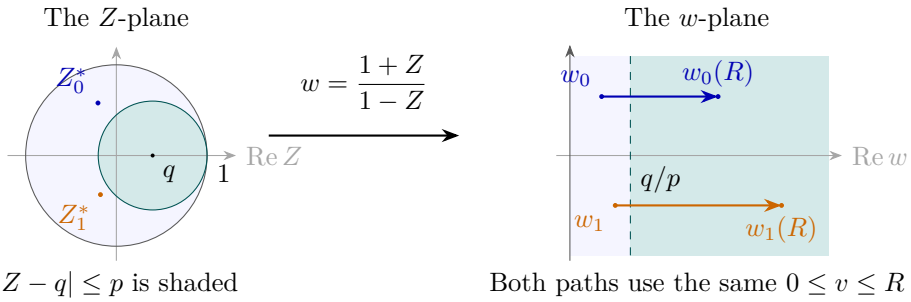


Figure 2: Geometry of the simultaneous continuation. The Cayley map turns the deletion disk into $\operatorname{Re} w \ge q/p$. Two possible starting coordinates are shown; the horizontal arrows have different lengths but share the same parameter: $w_r(v) = (1+v)\operatorname{Re} w_r + \mathrm{i}\operatorname{Im} w_r$ for $0 \le v \le R$. The proof interpolates $F(Z_0(v), \dots, Z_{k-1}(v))$ on one complex strip, simultaneously for every coordinate.

On the closed strip $-1/2 \le \operatorname{Re} v \le R$ we have $\operatorname{Re} w_r(v) = (1+\operatorname{Re} v)\xi_r > 0$, so all $Z_r(v)$ lie in the unit disk and depend analytically on v; at $v = 0$ they equal Z_r^*. On the right boundary $\operatorname{Re} w_r(v) = (1+R)\xi_r \ge q/p$, so (5.16) applies. Moreover

$$1 + R \le C\frac{Y^\alpha}{p}, \tag{5.19}$$

because $\gamma = c_2 q_* Y^{-\alpha}$ and $q \le q_*$.

We need a left-boundary bound free of n^k. The identity $1 - |Z|^2 = 4\operatorname{Re} w/|w+1|^2$ gives, at $\operatorname{Re} v = -1/2$,

$$1 - |Z_r(v)| \ge \frac{1 - |Z_r^*|}{4(1+|v|)^2}. \tag{5.20}$$

To check the constant, first use $1 - |Z_r(v)| \ge \xi_r/|w_r(v)+1|^2$ and $1 - |Z_r^*| \le 4\xi_r/|w_r+1|^2$, then $|w_r(v)+1| \le |w_r+1| + \xi_r|v| \le (1+|v|)|w_r+1|$. The coefficients of F have magnitude at most two, so summing independent geometric series over the positive prefix and gap lengths gives, in the open unit polydisk,

$$|F(Z)| \le 2\prod_{r=0}^{k-1} \frac{1}{1-|Z_r|}.$$

It follows from (5.20) that

$$g(v) := \frac{F(Z_0(v), \dots, Z_{k-1}(v))}{(2+v)^{2k}}$$

has the boundary bounds

$$\begin{aligned} &\log|g(v)| \le W_* + Ck + C \qquad (\operatorname{Re} v = -1/2), \\ &\log|g(v)| \le k\log(n/p) + \log\varepsilon \quad (\operatorname{Re} v = R). \end{aligned} \tag{5.21}$$

On the left $(1+|v|)/|2+v|$ is bounded absolutely; on the right $|2+v| \geq 2+R > 1$. The function g is analytic and bounded on the strip, since F is a finite polynomial on the unit polydisk and $|2+v| \geq 3/2$.

The strip estimate we use is a consequence of the maximum principle. If a bounded analytic function on $a \leq \operatorname{Re} v \leq b$ has logarithmic boundary bounds L_-, L_+, then at real $t \in [a,b]$ its logarithm is at most $((b-t)L_- + (t-a)L_+)/(b-a)$. Multiply by the exponential of the negative affine interpolant to make both sides one; on rectangles of height tending to infinity apply the maximum principle after multiplying by $\exp\{\tau(v^2 - \max\{a^2, b^2\})\}$ for $\tau > 0$, which has modulus at most one on the vertical sides and tends to zero on the horizontal sides. Let the height tend to infinity, then $\tau \to 0$; this proves the estimate, including zeros of the function by continuity.

At $t = 0$ the right boundary has weight $1/(2R+1)$. By (5.11), $\log|g(0)| \geq -E_h - 1 - 2k\log 2$. Substituting (5.21) gives

$$\begin{aligned} \log(1/\varepsilon) &\leq k\log(n/p) + (2R+1)(E_h + 1 + 2k\log 2) + 2R(W_* + Ck + C) \\ &\leq k\log(n/p) + C(1+R)(E_h + 4^h A_0). \end{aligned} \tag{5.22}$$

This is the quantitative transfer from input coordinates to trace coordinates.

It remains to compare this exponent with the budget (1.1). Put $t = \log_2 \mu$. The choices in (5.2) give $t + \alpha < 3$, and

$$4^h \leq CY^2, \qquad E_h \leq CA_0 Y^t, \qquad A_0 \leq Q(1+\log Y), \qquad QY \leq C\log n.$$

Thus (5.19) yields

$$\begin{aligned} (1+R)E_h &\leq \frac{C}{p}(\log n)Y^{t+\alpha-1}(1+\log Y) \leq \frac{C}{p}(\log n)Y^2(1+\log Y), \\ (1+R)4^h A_0 &\leq \frac{C}{p}(\log n)Y^{1+\alpha}(1+\log Y) \leq \frac{C}{p}(\log n)Y^2(1+\log Y). \end{aligned}$$

Also, since $n \geq 2$ and $\log(1/p) \leq 1/p$,

$$k\log(n/p) \leq CY^2\big(\log n + \log(1/p)\big) \leq \frac{C}{p}(\log n)Y^2.$$

Using (5.1), the right-hand side of (5.22) is at most

$$\frac{C}{p}(\log n)Y^2(1+\log Y) \leq C_6\, p^{-1}(\log n)\mathsf{X}(n,p)^2\big(1+\log \mathsf{X}(n,p)\big)^6 = C_6\, B(n,p). \tag{5.23}$$

Taking $\varepsilon = e^{-C_{\mathrm{sep}} B(n,p)}$ with $C_{\mathrm{sep}} > C_6$ contradicts (5.22), so (5.15) fails for that ε. This proves Theorem 3.2 with the cutoff (5.2).

All constants were fixed in this order: those of Proposition 4.1; then J_*, H, η; then C_5, c_2; and finally C_{sep}. Nothing depends on the denominator of p or on how close p is to an endpoint. In particular, when $q < 1/n$ the cutoff $q_* = 1/n$ keeps $Y < 2$ and $Q = 1 + \log n$, while $q/q_* \leq 1$ keeps the strip estimate uniform; when p is small the strip width is controlled by the single factor p^{-1} already in the budget. The case $p = 1$ is handled directly in the decoding argument.

6 A uniform decoder

Theorem 3.2 gives a test for the first wrong bit of a prefix: no moment system that fixes that prefix can match all the low-order trace statistics. To turn this into an algorithm we must also recognise

a *correct* prefix in bounded time. The extra fact that makes this possible is that a correct prefix admits a moment system with strict slack, both in its statistical constraints and in its positivity condition. We first exhibit that slack, then give a rational procedure that is guaranteed to find a feasible system whenever slack exists.

6.1 Statistical constraints for a prefix

Assume for now $p < 1$. Choose an integer D and a positive dyadic number $\zeta = 2^{-b_\zeta}$ so that

$$D_* \le D \le CY^2, \quad \zeta \le \tfrac{1}{4}, \quad 4\zeta < e^{-C_{\mathrm{sep}} B(n,p)}, \quad b_\zeta = O(B(n,p)), \tag{6.1}$$

with C_{sep} from Theorem 3.2. Integer formulas for these choices come later. Let

$$\mathcal{I}_D = \{I \subseteq \{1,\dots,n\} : 1 \le |I| \le D\}, \quad N = |\mathcal{I}_D|, \quad T = 512(1+N)\zeta^{-2},$$

which is an integer. From T independent traces form the empirical mean $\widehat{t}_I$ of the rank observable defining T_I, for every $I \in \mathcal{I}_D$. These are ordinary trace observables: a requested rank that is absent contributes zero.

For each fixed input x let $\mathcal{G}$ be the event

$$|\widehat{t}_I - T_I(x)| \le \zeta/8 \quad \text{for every } I \in \mathcal{I}_D. \tag{6.2}$$

Each observable takes values in $\{0,1\}$, so its empirical mean has variance at most $1/(4T)$. Chebyshev and a union bound give

$$\mathbb{P}_x(\mathcal{G}^c) \le \frac{16N}{T\zeta^2} \le \frac{1}{32}. \tag{6.3}$$

The same event controls all the prefix tests, even though the prefixes are chosen using the data.

For a proposed prefix, use the remaining sign variables and their moment matrix $M(m) = \mathcal{M}_D(m)$ from (3.3); its nonempty moment coordinates form a real vector m. Consider the rational constraints

$$m \in [-1,1]^l, \quad M(m) \succeq 0, \quad |\mathcal{E}_m[T_I] - \widehat{t}_I| \le \zeta \quad (I \in \mathcal{I}_D), \tag{6.4}$$

where l is the number of moment coordinates. Substituting the prefix and writing each remaining bit as one half of one plus its sign makes $\mathcal{E}_m[T_I]$ an affine function of m; its nonconstant coefficients have absolute sum at most one by (3.2).

Lemma 6.1. *On $\mathcal{G}$, a prefix whose first incorrect bit is at its last position has no solution to (6.4). Every correct prefix has a moment vector $m^* \in [-1,1]^l$ such that, with G the dimension of its moment matrix,*

$$M(m^*) \succeq (\zeta/8)\,\mathrm{Id}_G, \qquad |\mathcal{E}_{m^*}[T_I] - \widehat{t}_I| \le \zeta/2 \quad (I \in \mathcal{I}_D). \tag{6.5}$$

Proof. If a wrong prefix had a feasible moment vector, then (6.2) and (6.4) would give $|\mathcal{E}_m[T_I] - T_I(x)| \le 9\zeta/8$ for all $I \in \mathcal{I}_D$, contradicting Theorem 3.2 and (6.1).

For a correct prefix, mix the moments of the true completion with weight $1-\zeta/8$ and the moments of independent uniform signs on the remaining positions with weight $\zeta/8$. The latter moment matrix is the identity: its (I,J) entry is zero unless $I = J$. The mixture has the stated matrix lower bound and coordinates in $[-1,1]$. Both constituent distributions give channel expectations in $[0,1]$, so mixing changes each $T_I(x)$ by at most $\zeta/8$; (6.2) then bounds the discrepancy from $\widehat{t}_I$ by $\zeta/4$, stronger than required. □

6.2 Finding a moment system with slack

The next lemma gives the bounded procedure for (6.4). It only promises to succeed when there is slack, and every reported solution satisfies the exact inequalities. This sidesteps deciding arbitrary semidefinite feasibility. The procedure is a projected relaxation method in the tradition of Agmon and Motzkin–Schoenberg [1, 16]; we give the iteration and its rounding analysis explicitly.

Lemma 6.2 (Rational feasibility with slack)**.** *Let $r \ge 0$ and $D \ge 1$ be integers, and let m consist of the*

$$l = \sum_{j=1}^{\min(2D,r)} \binom{r}{j}$$

nonempty sign moments. Let $M(m)$ be their moment matrix, of dimension

$$G = \sum_{j=0}^{\min(D,r)} \binom{r}{j}.$$

Suppose a finite list of rational affine inequalities $\ell_t(m) \le b_t$ is given, each with nonconstant coefficient vector of ℓ^1-norm at most one, together with a dyadic $0 < \zeta \le 1/4$. Then a deterministic rational procedure either returns a vector with

$$m \in [-1,1]^l, \quad M(m) \succeq 0, \quad \ell_t(m) \le b_t \text{ for all } t,$$

or reports failure. It returns a vector whenever some $m^* \in [-1,1]^l$ satisfies

$$M(m^*) \succeq (\zeta/8)\,\mathrm{Id}_G, \quad \ell_t(m^*) \le b_t - \zeta/2 \text{ for all } t.$$

Put $\lambda = \zeta/(16G)$. The procedure performs at most $\lceil 10(1+l)/\lambda^2 \rceil$ feasibility checks, each taking polynomially many bit operations in the list dimensions, the input bit lengths, and $\log(1/\zeta)$. These time bounds hold without the slack assumption.

Proof. If $l = 0$ the matrix is [1] and all inequalities are constant, so direct rational comparisons suffice. Suppose $l > 0$. We give exact separating directions, then an iteration with bounded precision.

Directions from violated constraints. At a current point $m^{\mathrm{cur}} \in [-1,1]^l$ first check the affine inequalities and the matrix condition exactly. If all hold, return the current vector. A violated inequality $\ell_t(m^{\mathrm{cur}}) > b_t$ gives the direction a equal to minus its nonconstant coefficient vector. Then $\|a\|_2 \le 1$ and, under the slack assumption, $\langle m^* - m^{\mathrm{cur}}, a\rangle \ge \zeta/2 \ge \lambda$.

If instead the matrix condition fails, find a nonzero rational vector v with $v^\top M(m^{\mathrm{cur}})v < 0$ (constructed just below). Write the associated affine form as $v^\top M(m)v = c + h \cdot m$ and set $a = h/(G\,v^\top v)$. Each matrix entry is the constant 1 or one moment coordinate, so $\|h\|_1 \le \|v\|_1^2 \le G\|v\|_2^2$ and $\|a\|_2 \le 1$. The slack matrix bound gives

$$\langle m^* - m^{\mathrm{cur}}, a\rangle = \frac{v^\top(M(m^*) - M(m^{\mathrm{cur}}))v}{G\,v^\top v} \ge \frac{\zeta}{8G} \ge \lambda.$$

Thus either violated condition yields a rational direction with

$$\|a\|_2 \le 1, \qquad \langle m^* - m^{\mathrm{cur}}, a\rangle \ge \lambda \tag{6.6}$$

whenever a slack solution exists. A violated constant inequality has zero direction; that cannot happen in the slack case, and otherwise the iteration is still bounded.

An exact matrix check. For a rational symmetric matrix, eliminate a positive diagonal pivot via its Schur complement. Congruence by an invertible triangular matrix shows the original is positive semidefinite exactly when the positive pivot and this Schur complement are. A negative diagonal gives a negative unit vector. If a remaining diagonal at i is zero and its entry b in column j is nonzero, let c be the diagonal at j; we may assume $c \geq 0$, else a negative diagonal already exists. Then

$$y = e_i - \frac{b}{1 + c + |b|} e_j$$

has quadratic form $\frac{b^2}{(1+c+|b|)^2}\big[-2(1 + c + |b|) + c\big] < 0$. Otherwise continue with a positive diagonal pivot; when none remain, all entries are zero and the remainder is positive semidefinite. A negative vector of a Schur complement lifts to one of the original matrix using only rational arithmetic: with P the eliminated block and B its block against the remaining indices, the lift of y is $v = (-P^{-1}By,\, y)$, whose quadratic form equals that of y for the Schur complement.

These operations have polynomial bit complexity. Clear denominators in a rational matrix with t-bit entries; the product of all denominators has at most $G^2 t$ bits. If H is the resulting integer matrix with s-bit entries, the determinant expansion bounds every j-row minor by $j!\, 2^{js}$, of bit length $O(G(s + \log G))$. Schur-complement entries are ratios of minors, and the inverse formula for P gives the same bound for the lift. Fraction-reduced elimination with rational comparisons computes the check and the negative vector in polynomially many bit operations, however small a positive pivot or negative quadratic form may be.

Iteration and rounding. Start with $m^{\mathrm{cur}} = 0$. Choose the dyadic grid with mesh

$$\eta = 2^{-g}, \qquad g = \left\lceil \log_2 \frac{100(1 + l)}{\lambda^2} \right\rceil.$$

The integer g is computed by rational comparisons with powers of two, so $\eta \leq \lambda^2/(100(1 + l))$, with $O(1 + \log((1 + l)/\lambda))$ fractional bits. After an unsuccessful check, add λa, clip every coordinate to $[-1, 1]$, and round to the nearest grid point in the box (both box endpoints lie on the grid). Repeat up to the stated number of checks, reporting failure if none accepts.

Under the slack assumption (6.6) gives

$$\|m^{\mathrm{cur}} + \lambda a - m^*\|_2^2 \leq \|m^{\mathrm{cur}} - m^*\|_2^2 - \lambda^2.$$

Clipping cannot increase distance to m^*; since both the clipped vector and m^* lie in the box, rounding changes the squared distance by at most $4l\eta + l\eta^2 < \lambda^2/2$. Each failed check thus decreases squared distance by at least $\lambda^2/2$, whose initial value is at most l, so acceptance occurs before the cap. Finally, rounding keeps every iterate on a single grid of the indicated bit length; the exact matrix check, the affine comparisons, forming and normalizing h, and one update involve polynomially many rational operations on numbers of polynomial bit length, and temporary denominators do not accumulate across updates. The fixed cap and these bounds apply to every rational input list, with or without a slack solution. □

For a prefix test, write each absolute-value constraint of (6.4) as two affine inequalities and apply Lemma 6.2. Lemma 6.1 verifies exactly the slack hypotheses for a correct prefix, and every acceptance, on any data, gives an actual moment system satisfying (6.4).

The decoder starts with the empty prefix and tests both one-bit extensions. If precisely one accepts, it appends that bit; otherwise it appends 0. It repeats until the prefix has length n. The latter rule specifies a length-n output even on data where the statistical estimates fail. There are at most $2n$ tests.

6.3 Integer parameters and bit complexity

We now realise the choices in (6.1) using only the binary input. Continue to assume $p < 1$ and use q_*, Q, Y from (3.7). Define the positive integers

$$u = 1 + \lceil \log_2 n \rceil, \qquad v = 1 + \lfloor \log_2(1/q_*) \rfloor,$$
$$y = 1 + \lceil u/v \rceil, \qquad w = 1 + \lceil \log_2 y \rceil, \qquad a_p = \lceil 1/p \rceil.$$

All logarithms here mean integer length comparisons. In particular the integer part of a logarithm of a rational number is found by comparing its numerator with power-of-two multiples of its denominator, and the rational maximum defining q_* is found by an integer comparison.

The elementary comparisons

$$u \asymp \log n, \quad v \asymp Q, \quad Y \le y \le CY, \quad w \asymp 1 + \log Y, \quad p^{-1} \le a_p \le 2p^{-1}$$

have absolute constants. For the bound on y use $u \ge (\log n)/\log 2$ and $v \le Q/\log 2$. With $Y \asymp \mathsf{X}(n,p)$ these give

$$B(n,p) \le a_p\, u\, y^2\, w^6 \le C\, B(n,p). \tag{6.7}$$

Choose fixed positive integers K_D, K_m large enough for the constants in Theorem 3.2 and set

$$D = K_D y^2, \qquad b_\zeta = K_m\, a_p\, u\, y^2\, w^6, \qquad \zeta = 2^{-b_\zeta}. \tag{6.8}$$

Since $D_* \le CY^2$ and $B(n,p) \ge \log 2$, these ensure all of (6.1). The integers D, b_ζ in binary are computable in time polynomial in the input bit lengths; writing the denominator 2^{b_ζ} then costs $O(b_\zeta)$ bits, included below.

Next form the rational constraints. For $I = \{l_1 < \cdots < l_k\}$ and $i_1 < \cdots < i_k$ the coefficient of $X_{i_1} \cdots X_{i_k}$ in T_I is

$$\binom{i_1 - 1}{l_1 - 1} \prod_{j=1}^{k-1} \binom{i_{j+1} - i_j - 1}{l_{j+1} - l_j - 1} p^{l_k} q^{i_k - l_k}, \tag{6.9}$$

when all gap counts are possible and zero otherwise. Indeed the selected positions must be retained, and the prefix and successive open gaps must contain the indicated numbers of extra retentions; positions after i_k are unrestricted. Formula (6.9) follows from multiplying these independent binomial probabilities.

If the input representation is $p = a/b$, every denominator in (6.9) divides b^n. Binomial coefficients have $O(n)$ bits. Sign expansion adds a denominator dividing 2^D, and empirical means have denominator T. Hence the constraint coefficients have polynomial bit length in $n + L + D + \log T$. Enumerating the rank and input tuples, expanding the resulting bit monomials, and substituting the prefix constructs them directly; no list of complete trace masks is needed.

These enumerations fit the resource scale. From $D = O(Y^2)$, $Y \asymp \mathsf{X}(n,p)$, and (1.1),

$$D \log n = O(B(n,p)). \tag{6.10}$$

The rank list, every matrix basis, and every moment list have size at most $(n+1)^{2D} = \exp(O(B(n,p)))$, and so do input tuple lists. A sign expansion has at most 2^D terms; products of these enumerations remain $\exp(O(B(n,p)))$. Subsets larger than the available positions are simply absent, so this includes $D > n$ and prefixes with very few remaining positions.

By (6.1), T is $\exp(O(B(n,p)))$. For every prefix the iteration cap of Lemma 6.2 is at most

$$2560(1 + l)G^2\zeta^{-2} + 1 = \exp(O(B(n,p))),$$

and the bit length of its grid is $O(b_\zeta + \log G + \log(1+l)) = O(B(n,p))$. The coefficient bounds and that lemma bound all computation by a fixed polynomial in

$$n + L + \exp(O(B(n,p))), \tag{6.11}$$

including parameter and constraint construction, exact rational arithmetic, and the at most $2n$ prefix tests.

Proof of Theorem 1.1. If $p = 1$, request one trace and output it. For the universal halting specification, output 0^n if the supplied well-formed trace has a different length; the actual channel always supplies the input itself, and this branch takes polynomially many bit operations in $n + L$.

For $p < 1$, use (6.8), request the T traces, and compute the empirical statistics. Run the prefix rule. This defines the machine on every stream of well-formed traces.

On $\mathcal{G}$, induction on prefix length shows precisely the correct extension accepts: the incorrect extension is infeasible by Lemma 6.1, and the correct one has the slack that guarantees acceptance. Thus the output is x on $\mathcal{G}$, whose probability is at least $31/32 > 2/3$ for each fixed x.

Each trace has at most n bits and one delimiter. Reading the traces and computing each of the finitely many rank observables costs polynomially many bit operations in $n + T + N$; the empirical counters have $O(\log T)$ bits. All loops, including all feasibility tests, have the input-dependent caps already given. Thus (6.11) bounds every well-formed trace stream, independently of the success event, and the machine can ignore its internal random bits.

To pin the computational model, the finite lists are stored sequentially on tapes; replacing an indexed lookup by a scan costs at most a polynomial factor in their total encoded length. Integer addition, multiplication, division, and fraction reduction have elementary polynomial-time multitape implementations. The bit-length and loop bounds thus give absolute constants K, d with trace count at most $\exp(KB(n,p))$ and total bit cost at most $(n + L + \exp(KB(n,p)))^d$, after enlarging K, d to absorb fixed factors. This enlargement is uniform because $B(n,p) \geq \log 2$. Choose absolute C, c with $C \geq K$ and $c \geq d$; both the sample bound $M_C(n,p)$ and the runtime $(n + L + M_C(n,p))^c$ follow. All constants in parameter selection are fixed integers in one finite program. □

References

[1] Shmuel Agmon. The relaxation method for linear inequalities. *Canadian Journal of Mathematics* **6** (1954), 382–392.

[2] Tuğkan Batu, Sampath Kannan, Sanjeev Khanna, and Andrew McGregor. Reconstructing strings from random traces. In *Proceedings of the Fifteenth Annual ACM–SIAM Symposium on Discrete Algorithms (SODA)*, 2004, pp. 910–918.

[3] Peter Borwein and Tamás Erdélyi. Littlewood-type problems on subarcs of the unit circle. *Indiana University Mathematics Journal* **46**(4) (1997), 1323–1346.

[4] Arnav Burudgunte, Paul Valiant, and Hongao Wang. Quasipolynomial trace reconstruction. Preprint, 2026. arXiv:2607.04073v1.

[5] Zachary Chase. New lower bounds for trace reconstruction. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* **57**(2) (2021), 627–643.

[6] Zachary Chase. Separating words and trace reconstruction. In *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, 2021, pp. 21–31.

[7] Xi Chen, Anindya De, Chin Ho Lee, and Rocco A. Servedio. Trace reconstruction from local statistical queries. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2024)*, LIPIcs **317** (2024), 52:1–52:24.

[8] Xi Chen, Anindya De, Chin Ho Lee, Rocco A. Servedio, and Sandip Sinha. Polynomial-time trace reconstruction in the low deletion rate regime. In *12th Innovations in Theoretical Computer Science Conference (ITCS 2021)*, LIPIcs **185** (2021), 20:1–20:20.

[9] Xi Chen, Anindya De, Chin Ho Lee, Rocco A. Servedio, and Sandip Sinha. Polynomial-time trace reconstruction in the smoothed complexity model. In *Proceedings of the 2021 ACM–SIAM Symposium on Discrete Algorithms (SODA)*, 2021, pp. 54–73. arXiv:2008.12386.

[10] Kuan Cheng, Elena Grigorescu, Xin Li, Madhu Sudan, and Minshen Zhu. On k-mer-based and maximum likelihood estimation algorithms for trace reconstruction. In *2024 IEEE International Symposium on Information Theory (ISIT)*, 2024, pp. 879–884. arXiv:2308.14993v2.

[11] Anindya De, Ryan O'Donnell, and Rocco A. Servedio. Optimal mean-based algorithms for trace reconstruction. *Annals of Applied Probability* **29**(2) (2019), 851–874.

[12] Nina Holden and Russell Lyons. Lower bounds for trace reconstruction. *Annals of Applied Probability* **30**(2) (2020), 503–525.

[13] Thomas Holenstein, Michael Mitzenmacher, Rina Panigrahy, and Udi Wieder. Trace reconstruction with constant deletion probability and related results. In *Proceedings of the Nineteenth Annual ACM–SIAM Symposium on Discrete Algorithms (SODA)*, 2008, pp. 389–398.

[14] Jean B. Lasserre. Global optimization with polynomials and the problem of moments. *SIAM Journal on Optimization* **11**(3) (2001), 796–817.

[15] Vladimir I. Levenshtein. Efficient reconstruction of sequences. *IEEE Transactions on Information Theory* **47**(1) (2001), 2–22.

[16] Theodore S. Motzkin and Isaac J. Schoenberg. The relaxation method for linear inequalities. *Canadian Journal of Mathematics* **6** (1954), 393–404.

[17] Fedor Nazarov and Yuval Peres. Trace reconstruction with $\exp(O(n^{1/3}))$ samples. In *Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, 2017, pp. 1042–1046.

[18] Pablo A. Parrilo. *Structured semidefinite programs and semialgebraic geometry methods in robustness and optimization*. Ph.D. thesis, California Institute of Technology, 2000.